Reg. No. | | | | | | | | | | |

[18MSCH3S32]

THIRD SEMESTER UNIVERSITY EXAMINATION, DECEMBER 2021

PHARMACEUTICAL AND INDUSTRIAL CHEMISTRY

Time : 03 Hours

Maximum Marks: 70

Instructions:
1. Question 1 is compulsory, carries 10 marks.
2. Questions 2 to 7 carries 15 marks each, answer any FOUR.

1. Answer any FIVE of the following questions. 5 x 2 = 10

a. Write a note on computer-aided drug design.

b. What are analgesic and antipyretic drugs? Write the structures of any two analgesic-antipyretics drug.

c. How do you synthesis diazenon from isobutyronitrile?

d. What are additives in paints?

e. Explain an anionic polymerization with examples.

f. List out the various elements present in petroleum and its concentration ranges.

2. a. Write a note on receptor based approach for pharmacophore identification.

b. Explain optimization of drug-target interaction on the basis of variation of substituents and simplification of the structure.

c. What are isosteres and bioisosteres? Explain with suitable examples. (5+5+5)

3. a. Explain the synthesis of acyclovir.

b. Write notes on the synthesis of sorbitrate and methyldopa.

c. What are antimalarial drugs? Outline the synthesis and SAR of chloroquine. (4+5+6)

4. a. Explain the synthesis of carbofuran and aldicarb.

b. Write a note on mode of action of carbamate insecticides.

c. Discuss the classification of pheromone. (6+4+5)

5. a. What are dyes? Explain its composition and classification.

b. Write a note on polyurethane resins. (10+5)

6. a. Explain i) emulsion polymerization and its advantages.

ii) solution polymerization and its limitations.

b. How is cationic polymerization different from anionic polymerization?

c. What is compounding of resin? Explain. (5+5+5)

7. a. Describe the biogenesis and abiogenesis origin of petroleum.

b. Write a note on the various stages of the refining process. (10+5)

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112